STATISTICS

Chapter 8 Homework

Confidence intervals and their interpretation

Sections 8.1-8.3 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Means: choose the right uncertainty

1. An independent random sample of 64 package masses has mean 50 g. The population SD is known to be 8 g. Use a normal-model 95% interval with critical value 1.96. Find the standard error, margin of error, and interval; state the relevant sampling/normal-approximation conditions.

2. Interpret the confidence level from Problem 1 in terms of repeated samples. Explain why the interval does not claim that 95% of individual packages have masses in this interval.

3. A separate independent random sample of 16 masses from a normal population has mean 50 g and sample SD 8 g. The population SD is unknown. Use t* = 2.131 for 95% confidence. Find degrees of freedom, standard error, margin of error, and the interval. Why use t?

4. For the known-SD setting in Problem 1, explain how interval width changes if (a) confidence increases with n unchanged or (b) n increases from 64 to 256 at the same confidence. Compute the new standard error for (b).

Proportions and planning

5. In an independent random sample of 100 students, 60 prefer format A. Check the textbook success/failure counts (> 5 each) for an ordinary normal-approximation interval. Compute a 95% interval using z* = 1.96 and standard error √[p-hat(1 - p-hat)/n] ≈ 0.0490.

6. Explain what the interval in Problem 5 suggests about whether more than half of all students prefer A. Does it establish that format A causes better learning?

7. For a future 95% proportion interval with margin of error at most 0.05, use the conservative planning value p = 0.5 and the formula below. Compute n and explain how to round it.

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2}$$

8. Choose the method: known-SD z interval, unknown-SD t interval, or proportion interval. (a) Estimate mean time with population SD known. (b) Estimate mean time with only sample SD. (c) Estimate the fraction that chooses A. Name the population parameter each targets.

9. A sample of 8 from a strongly skewed population has an extreme outlier and unknown population SD. Explain why a routine t interval may be unreliable. Would a narrow interval remove the concern?

10. A website uses a voluntary poll of 5,000 visitors and reports a very narrow interval for all college students. Explain what the narrow interval fails to address and propose a better sampling design.